

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Y.POOJA SRI	Reg. No.:	192324143
Section / Batch:	AIDS-2023	Date:	

Code of Conduct

I _____ Y.POOJA SRI / 192324143 _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Y.POOJA SRI _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging, HMM	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q1

Annexure A – SCENARIO BASED

A company is developing an NLP-based grammar checker for educational applications. The system is trained using a manually POS-tagged corpus. Your task is to recreate the Hidden Markov Model (HMM) **without using any built-in NLP or HMM libraries**.

From the given corpus,

- determine the **Emission Probability**
- determine the **Transition Probability**
- implement the **Viterbi Algorithm**
- predict the POS tags for the new input sentence.

Use only Python dictionaries, loops, and basic programming constructs.

Training Corpus

Sentence	POS Tags
The/DT boy/NN eats/VBZ rice/NN	DT NN VBZ NN
The/DT girl/NN drinks/VBZ milk/NN	DT NN VBZ NN
A/DT cat/NN drinks/VBZ milk/NN	DT NN VBZ NN
The/DT dog/NN chases/VBZ cat/NN	DT NN VBZ NN
A/DT teacher/NN teaches/VBZ students/NNS	DT NN VBZ NNS
Students/NNS study/VBP English/NN	NNS VBP NN
Birds/NNS fly/VBP high/RB	NNS VBP RB
Children/NNS play/VBP games/NNS	NNS VBP NNS

New Input Sentence

The cat drinks milk

Tasks

1. Separate each sentence into words and POS tags.
2. Compute the **Emission Probability** of every word.
3. Compute the **Transition Probability** between POS tags.
4. Recreate the Hidden Markov Model using Python dictionaries.
5. Implement the **Viterbi Algorithm** without using any built-in HMM/NLP libraries.
6. Predict the POS tag sequence for "The cat drinks milk"

Program:

```
corpus = [  
    ("The boy eats rice".split(),["DT","NN","VBZ","NN"]),  
    ("The girl drinks milk".split(),["DT","NN","VBZ","NN"]),  
    ("A cat drinks milk".split(),["DT","NN","VBZ","NN"]),  
    ("The dog chases cat".split(),["DT","NN","VBZ","NN"]),  
    ("A teacher teaches students".split(),["DT","NN","VBZ","NNS"]),  
    ("Students study English".split(),["NNS","VBP","NN"]),  
    ("Birds fly high".split(),["NNS","VBP","RB"]),  
    ("Children play games".split(),["NNS","VBP","NNS"])  
]
```

```
print("Training Corpus\n")  
for words, tags in corpus:  
    print("Words :", words)  
    print("Tags :", tags)  
    print()
```

```
emission_count={}  
tag_count={}  
for words, tags in corpus:  
    for word, tag in zip(words, tags):  
        if tag not in emission_count:  
            emission_count[tag]={}  
        emission_count[tag][word]=emission_count[tag].get(word,0)+1
```

```

        tag_count[tag]=tag_count.get(tag,0)+1
print("\nEmission Probability\n")
emission_prob={}
for tag in emission_count:
    emission_prob[tag]={}
    for word in emission_count[tag]:
        emission_prob[tag][word]=emission_count[tag][word]/tag_count[tag]
        print(word,"|",tag,"=",
              round(emission_prob[tag][word],3))

transition_count={}
transition_total={}
for words,tags in corpus:
    for i in range(len(tags)-1):
        t1=tags[i]
        t2=tags[i+1]
        if t1 not in transition_count:
            transition_count[t1]={}
        transition_count[t1][t2]=transition_count[t1].get(t2,0)+1
        transition_total[t1]=transition_total.get(t1,0)+1
print("\nTransition Probability\n")
transition_prob={}
for tag in transition_count:
    transition_prob[tag]={}
    for next_tag in transition_count[tag]:
        transition_prob[tag][next_tag]=transition_count[tag][next_tag]/transition_total[tag]
        print(tag,"->",next_tag,"=",
              round(transition_prob[tag][next_tag],3))

```

```

HMM={
"Emission":emission_prob,
"Transition":transition_prob
}
sentence="The cat drinks milk".split()
states=list(emission_prob.keys())
viterbi=[]
previous={}

for word in sentence:
    scores={}
    for state in states:
        emission=emission_prob.get(state,{}).get(word,0)
        if not viterbi:
            scores[state]=emission
        else:
            best=0
            for prev in states:
                transition=transition_prob.get(prev,{}).get(state,0)
                score=viterbi[-1][prev]*transition*emission
                if score>best:
                    best=score
            scores[state]=best
    viterbi.append(scores)

```

```

print("\nPredicted POS Tags\n")
predicted=[]
for score in viterbi:
    best=max(score,key=score.get)
    predicted.append(best)

```

```
for w,t in zip(sentence,predicted):
```

```
print(w,"--> ",t)
```

Output:

```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
===== RESTART: C:/Users/Pooja Sri/Downloads/NLP1.py =====

Training Corpus

Words : ['The', 'boy', 'eats', 'rice']
Tags  : ['DT', 'NN', 'VBZ', 'NN']

Words : ['The', 'girl', 'drinks', 'milk']
Tags  : ['DT', 'NN', 'VBZ', 'NN']

Words : ['A', 'cat', 'drinks', 'milk']
Tags  : ['DT', 'NN', 'VBZ', 'NN']

Words : ['The', 'dog', 'chases', 'cat']
Tags  : ['DT', 'NN', 'VBZ', 'NN']

Words : ['A', 'teacher', 'teaches', 'students']
Tags  : ['DT', 'NN', 'VBZ', 'NNS']

Words : ['Students', 'study', 'English']
Tags  : ['NNS', 'VBP', 'NN']

Words : ['Birds', 'fly', 'high']
Tags  : ['NNS', 'VBP', 'RB']

Words : ['Children', 'play', 'games']
Tags  : ['NNS', 'VBP', 'NNS']

Emission Probability

The | DT = 0.6
A   | DT = 0.4
boy | NN = 0.1
rice | NN = 0.1
girl | NN = 0.1
milk | NN = 0.2
cat  | NN = 0.2
dog  | NN = 0.1
teacher | NN = 0.1
English | NN = 0.1
eats | VBZ = 0.2
drinks | VBZ = 0.4
chases | VBZ = 0.2
teaches | VBZ = 0.2
students | NNS = 0.2
Students | NNS = 0.2
Birds | NNS = 0.2
Children | NNS = 0.2
games | NNS = 0.2
study | VBP = 0.333
fly | VBP = 0.333
play | VBP = 0.333
high | RB = 1.0

Transition Probability

DT -> NN = 1.0
NN -> VBZ = 1.0
VBZ -> NN = 0.8
VBZ -> NNS = 0.2
NNS -> VBP = 1.0
VBP -> NN = 0.333
VBP -> RB = 0.333
VBP -> NNS = 0.333

Predicted POS Tags

The --> DT
cat --> NN
drinks --> VBZ
milk --> NN
>>>
```

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____